WORKED SOLUTION

STEP I 1992, Question 11

Mechanics • Elastic Strings Equilibrium

> **STEP I 1992, Question 11**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/92-s1-q11

The key idea

Each particle is in equilibrium, so the tension just above it exceeds the tension just below it by mg. That links the three tensions, and Hooke's law turns tensions into extensions. The one extra fact we need is geometric: the three stretched lengths must add up to the height of the room. Once the heights are known, "$CD = a$" just says the lowest string carries no tension.

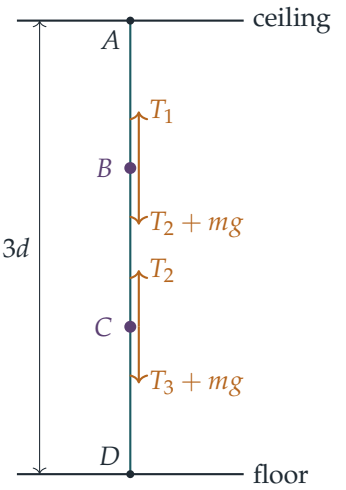
Solution

The model

The strings are light and elastic, obeying Hooke's law (tension $= \lambda \times$ extension$/a$) when stretched; the particles are at rest. Write T_1, T_2, T_3 for the tensions in AB, BC, CD and x_1, x_2, x_3 for their extensions, so $T_i = \lambda x_i / a$. It is convenient to name the extension that a weight mg alone would produce:

$$e = \frac{mga}{\lambda}.$$

We first assume all three strings are taut, and check afterwards when this is true. Measure heights y_B, y_C above the floor.



Heights of B and C

Resolving vertically for each particle (the string above pulls up, the string below and the weight pull down):

$$B: \quad T_1 = T_2 + mg, \qquad C: \quad T_2 = T_3 + mg.$$

Dividing by λ/a turns these into statements about extensions:

$$x_1 = x_2 + e, \qquad x_2 = x_3 + e, \qquad \text{so} \qquad x_2 = x_3 + e, \quad x_1 = x_3 + 2e.$$

The total length of the three strings is the height of the room:

$$3a + x_1 + x_2 + x_3 = 3d \quad \Longrightarrow \quad 3x_3 + 3e = 3d - 3a \quad \Longrightarrow \quad x_3 = d - a - e.$$

Then $x_2 = d - a$ and $x_1 = d - a + e$. The heights follow from the lengths of the lower strings:

$$y_C = CD = a + x_3 = d - e = d - \frac{mga}{\lambda},$$
$$y_B = y_C + BC = (d - e) + (a + d - a) = 2d - \frac{mga}{\lambda}.$$

As a check on AB: $3d - y_B = d + e = a + x_1$, as it should be.

Ask yourself When are these answers actually valid?

We assumed every string is taut, so we need every extension to be at least zero. The tightest condition is on the lowest string: $x_3 \geq 0$, that is

$$\frac{mga}{\lambda} \leq d - a.$$

(Then $x_2 = d - a \geq 0$ and $x_1 > 0$ automatically.) It makes sense that CD is the first to go slack: it supports nothing, and only the stretching of the room holds it taut. If

$mga/\lambda > d - a$, the string CD is slack and the system is just two particles hanging from A: then $T_2 = mg$, $T_1 = 2mg$, so $AB = a + 2e$ and $BC = a + e$, giving $y_B = 3d - a - 2e$ and $y_C = 3d - 2a - 3e$, provided this is still above the floor. The question intends the main case with all strings taut.

When can $CD = a$?

The string CD has its natural length exactly when it is just taut with no tension, $T_3 = 0$, i.e. $x_3 = 0$. This is the boundary between the two cases above, and either set of formulas gives

$$y_C = a \iff d - \frac{mga}{\lambda} = a \iff m = \frac{\lambda(d-a)}{ga}.$$

Physically, the particle at C is then held up entirely by BC, with $T_2 = mg$, and $T_1 = 2mg$.

A suitable mass exists exactly when this value of m is positive, which needs $d > a$. Conversely, if $d \leq a$ then no positive m works: if all strings were taut we would need $x_2 = d - a \geq 0$ and $x_3 = d - a - e < 0$, which is impossible, so CD is slack and C hangs at height $3d - 2a - 3e < 3a - 2a = a$ (or rests on the floor), so $CD < a$. Hence the set of values is

$$\{\, d : d > a \,\}.$$

For such d we get $y_C = a$, $y_B = d + a$ and $AB = 2d - a$.

Checks

- Units: mga/λ is a force times a length divided by a force, so a length, as it must be to subtract from d.
- With $m = 0$ the three strings share the stretch equally: each has length d, putting C at d and B at $2d$. Adding mass lowers both particles by the same amount e, since BC keeps extension $d - a$.
- Numerically, minimising the potential energy (gravitational plus elastic, with slack strings allowed) over $0 \leq y_C \leq y_B \leq 3d$ for 250 random choices of a, λ, d, m reproduced the heights in both the taut and the slack-CD cases to within 10^{-14}. With $a = 1$, $\lambda = 10$, $d = 2$ and $m = \lambda(d-a)/(ga) \approx 1.020$ it gave $CD = 1.000000$ and $y_B = 3.000000$; for $d \leq a$ no positive m gave $CD = a$.

Common slips

- Writing the tension as $\lambda\times$ (stretched length) $/a$ instead of $\lambda\times$ extension$/a$.
- Forgetting that the total length is fixed by the room: without $\sum(a + x_i) = 3d$ the equilibrium equations don't determine the extensions.
- Assuming every string must be in tension. Strings can go slack, and $CD = a$ happens precisely on the boundary of slackness.

Beyond STEP

This is a small statically indeterminate structure: equilibrium of the particles alone doesn't fix the tensions, and we needed a compatibility condition (the lengths add up) as well. The same idea, in matrix form, is how engineers analyse frameworks of springs and bars. Another way in is to minimise the total potential energy, which is how the numerical check worked; that variational view is central in later mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers springs, equilibrium and potential energy carefully.

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